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# Constrained RESTful Environments (CoRE) Link Format

## Abstract

This specification defines Web Linking using a link format for use by constrained web servers to describe hosted resources, their attributes, and other relationships between links. Based on the HTTP Link Header field defined in RFC 5988, the Constrained RESTful Environments (CoRE) Link Format is carried as a payload and is assigned an Internet media type. "RESTful" refers to the Representational State Transfer (REST) architecture. A well-known URI is defined as a default entry point for requesting the links hosted by a server.

## Status of This Memo

This is an Internet Standards Track document.

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## 1. Introduction

The Constrained RESTful Environments (CoRE) realizes the Representational State Transfer (REST) architecture [REST] in a suitable form for the most constrained nodes (e.g., 8-bit microcontrollers with limited memory) and networks (e.g., IPv6 over Low-Power Wireless Personal Area Networks (6LoWPANs) [RFC4919]). CoRE is aimed at Machine-to-Machine (M2M) applications such as smart energy and building automation.

The discovery of resources hosted by a constrained server is very important in machine-to-machine applications where there are no humans in the loop and static interfaces result in fragility. The discovery of resources provided by an HTTP [RFC2616] web server is typically called "Web Discovery" and the description of relations between resources is called "Web Linking" [RFC5988]. In the present specification, we refer to the discovery of resources hosted by a constrained web server, their attributes, and other resource relations as CoRE Resource Discovery.

The main function of such a discovery mechanism is to provide Universal Resource Identifiers (URIs, called links) for the resources hosted by the server, complemented by attributes about those resources and possible further link relations. In CoRE, this collection of links is carried as a resource of its own (as opposed to HTTP headers delivered with a specific resource). This document specifies a link format for use in CoRE Resource Discovery by extending the HTTP Link Header format [RFC5988] to describe these link descriptions. The CoRE Link Format is carried as a payload and is assigned an Internet media type. A well-known relative URI "/.well-known/core" is defined as a default entry point for requesting the list of links about resources hosted by a server and thus performing CoRE Resource Discovery. This specification is applicable for use with Constrained Application Protocol (CoAP) [COAP], HTTP, or any other suitable web transfer protocol. The link format can also be saved in file format.

### 1.1. Web Linking in CoRE

Technically, the CoRE Link Format is a serialization of a typed link as specified in [RFC5988], used to describe relationships between resources, so-called "Web Linking". In this specification, Web Linking is extended with specific constrained M2M attributes; links are carried as a message payload rather than in an HTTP Link Header field, and a default interface is defined to discover resources hosted by a server. This specification also defines a new relation

type "hosts" (from the verb "to host"), which indicates that the resource is hosted by the server from which the link document was requested.

In HTTP, the Link Header can be used to carry link information about a resource along with an HTTP response. This works well for the typical use case for a web server and browser, where further information about a particular resource is useful after accessing it. In CoRE, the main use case for Web Linking is the discovery of which resources a server hosts in the first place. Although some resources may have further links associated with them, this is expected to be an exception. For that reason, the CoRE Link Format serialization is carried as a resource representation of a well-known URI. The CoRE Link Format does reuse the format of the HTTP Link Header serialization defined in [RFC5988].

## 1.2. Use Cases

Typical use cases for Web Linking on today's web include, e.g., describing the author of a web page or describing relations between web pages (next chapter, previous chapter, etc.). Web Linking can also be applied to M2M applications, where typed links are used to assist a machine client in finding and understanding how to use resources on a server. In this section a few use cases are described for how the CoRE Link Format could be used in M2M applications. For further technical examples, see Section 5. As there is a large range of M2M applications, these use cases are purposely generic. This specification assumes that different deployments or application domains will define the appropriate REST Interface Descriptions along with Resource Types to make discovery meaningful.

### 1.2.1. Discovery

In M2M applications, for example, home or building automation, there is a need for local clients and servers to find and interact with each other without human intervention. The CoRE Link Format can be used by servers in such environments to enable Resource Discovery of the resources hosted by the server.

Resource Discovery can be performed either unicast or multicast. When a server's IP address is already known, either a priori or resolved via the Domain Name System (DNS) [RFC1034][RFC1035], unicast discovery is performed in order to locate the entry point to the resource of interest. In this specification, this is performed using a GET to "/.well-known/core" on the server, which returns a payload in the CoRE Link Format. A client would then match the appropriate Resource Type, Interface Description, and possible media type

[RFC2045] for its application. These attributes may also be included in the query string in order to filter the number of links returned in a response.

Multicast Resource Discovery is useful when a client needs to locate a resource within a limited scope, and that scope supports IP multicast. A GET request to the appropriate multicast address is made for "/.well-known/core". In order to limit the number and size of responses, a query string is recommended with the known attributes. Typically, a resource would be discovered based on its Resource Type and/or Interface Description, along with possible application-specific attributes.

### 1.2.2. Resource Collections

RESTful designs of M2M interfaces often make use of collections of resources. For example, an index of temperature sensors on a data collection node or a list of alarms on a home security controller. The CoRE Link Format can be used to make it possible to find the entry point to a collection and traverse its members. The entry point of a collection would always be included in "/.well-known/core" to enable its discovery. The members of the collection can be defined either through the Interface Description of the resource along with a parameter resource for the size of the collection or by using the link format to describe each resource in the collection. These links could be located under "/.well-known/core" or hosted, for example, in the root resource of the collection.

### 1.2.3. Resource Directory

In many deployment scenarios, for example, constrained networks with sleeping servers or large M2M deployments with bandwidth limited access networks, it makes sense to deploy resource directory entities that store links to resources stored on other servers. Think of this as a limited search engine for constrained M2M resources.

The CoRE Link Format can be used by a server to register resources with a resource directory or to allow a resource directory to poll for resources. Resource registration can be achieved by having each server POST their resources to "/.well-known/core" on the resource directory. This, in turn, adds links to the resource directory under an appropriate resource. These links can then be discovered by any client by making a request to a resource directory lookup interface.

### 1.3. Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this specification are to be interpreted as described in [RFC2119].

This specification makes use of the Augmented Backus-Naur Form (ABNF) [RFC5234] notation, including the core rules defined in Appendix B of that document.

This specification requires readers to be familiar with all the terms and concepts that are discussed in [RFC5988] and [RFC6454]. In addition, this specification makes use of the following terminology:

Web Linking
: A framework for indicating the relationships between web resources.

Link
: Also called "typed links" in [RFC5988]. A link is a typed connection between two resources identified by URI and is made up of a context URI, a link relation type, a target URI, and optional target attributes.

Link Format
: A particular serialization of typed links.

CoRE Link Format
: A particular serialization of typed links based on the HTTP Link Header field serialization defined in Section 5 of [RFC5988] but carried as a resource representation with a media type.

Attribute
: Properly called "Target Attribute" in [RFC5988]. A key/value pair that describes the link or its target.

CoRE Resource Discovery
: When a client discovers the list of resources hosted by a server, their attributes, and other link relations by accessing "/.well-known/core".

## 2. Link Format

The CoRE Link Format extends the HTTP Link Header field specified in [RFC5988]. The format does not require special XML or binary parsing, is fairly compact, and is extensible -- all important characteristics for CoRE. It should be noted that this link format is just one serialization of typed links defined in [RFC5988]; others

include HTML links, Atom feed links [RFC4287], or HTTP Link Header fields. It is expected that resources discovered in the CoRE Link Format may also be made available in alternative formats on the greater Internet. The CoRE Link Format is only expected to be supported in constrained networks and M2M systems.

Section 5 of [RFC5988] did not require an Internet media type for the defined link format, as it was defined to be carried in an HTTP header. This specification thus defines the Internet media type 'application/link-format' for the CoRE Link Format (see Section 7.3). Whereas the HTTP Link Header field depends on [RFC2616] for its encoding, the CoRE Link Format is encoded as UTF-8 [RFC3629]. A decoder of the format is not expected to validate UTF-8 encoding (but is not prohibited from doing so) and doesn't need to perform any UTF-8 normalization. UTF-8 data can be compared bitwise, which allows values to contain UTF-8 data without any added complexity for constrained nodes.

The CoRE Link Format is equivalent to the [RFC5988] link format; however, the ABNF in the present specification is repeated with improvements to be compliant with [RFC5234] and includes new link parameters. The link parameter "href" is reserved for use as a query parameter for filtering in this specification (see Section 4.1) and MUST NOT be defined as a link parameter. As in [RFC5988], multiple link descriptions are separated by commas. Note that commas can also occur in quoted strings and URIs but do not end a description. In order to convert an HTTP Link Header field to this link format, first the "Link:" HTTP header is removed, any linear whitespace (LWS) is removed, the header value is converted to UTF-8, and any percent-encodings are decoded.

```
 Link            = link-value-list
 link-value-list = [ link-value *[ "," link-value ]]
 link-value     = "<" URI-Reference ">" *( ";" link-param )
 link-param     = ( ( "rel" "=" relation-types )
                / ( "anchor" "=" DQUOTE URI-Reference DQUOTE )
                / ( "rev" "=" relation-types )
                / ( "hreflang" "=" Language-Tag )
                / ( "media" "=" ( MediaDesc
                       / ( DQUOTE MediaDesc DQUOTE ) ) )
                / ( "title" "=" quoted-string )
                / ( "title*" "=" ext-value )
                / ( "type" "=" ( media-type / quoted-mt ) )
                / ( "rt" "=" relation-types )
                / ( "if" "=" relation-types )
                / ( "sz" "=" cardinal )
                / ( link-extension ) )
 link-extension = ( parmname [ "=" ( ptoken / quoted-string ) ] )
```

```
                / ( ext-name-star "=" ext-value )
    ext-name-star  = parmname "*" ; reserved for RFC-2231-profiled
                                  ; extensions.  Whitespace NOT
                                  ; allowed in between.
    ptoken         = 1*ptokenchar
    ptokenchar     = "!" / "#" / "$" / "%" / "&" / "'" / "("
                   / ")" / "*" / "+" / "-" / "." / "/" / DIGIT
                   / ":" / "<" / "=" / ">" / "?" / "@" / ALPHA
                   / "[" / "]" / "^" / "_" / "`" / "{" / "|"
                   / "}" / "~"
    media-type     = type-name "/" subtype-name
    quoted-mt      = DQUOTE media-type DQUOTE
    relation-types = relation-type
                   / DQUOTE relation-type *( 1*SP relation-type ) DQUOTE
    relation-type  = reg-rel-type / ext-rel-type
    reg-rel-type   = LOALPHA *( LOALPHA / DIGIT / "." / "-" )
    ext-rel-type   = URI
    cardinal       = "0" / ( %x31-39 *DIGIT )
    LOALPHA        = %x61-7A   ; a-z
    quoted-string  = <defined in [RFC2616]>
    URI            = <defined in [RFC3986]>
    URI-Reference  = <defined in [RFC3986]>
    type-name      = <defined in [RFC4288]>
    subtype-name   = <defined in [RFC4288]>
    MediaDesc      = <defined in [W3C.HTML.4.01]>
    Language-Tag   = <defined in [RFC5646]>
    ext-value      = <defined in [RFC5987]>
    parmname       = <defined in [RFC5987]>
```

### 2.1. Target and Context URIs

Each link conveys one target URI as a URI-reference inside angle brackets ("<>"). The context URI of a link (also called the base URI in [RFC3986]) is determined by the following rules in this specification:

(a) The context URI is set to the anchor parameter, when specified.

(b) Origin of the target URI, when specified.

(c) Origin of the link format resource's base URI.

### 2.2. Link Relations

Since links in the CoRE Link Format are typically used to describe resources hosted by a server, the new relation type "hosts" is assumed in the absence of the relation parameter (see Section 7.2). The "hosts" relation type (from the verb "to host") indicates that

the target URI is a resource hosted by the server (i.e., server hosts resource) indicated by the context URI. The target URI MUST be a relative URI of the context URI for this relation type.

To express other relations, links can make use of any registered relation by including the relation parameter. The context of a relation can be defined using the anchor parameter. In this way, relations between resources hosted on a server or between hosted resources and external resources can be expressed.

### 2.3. Use of Anchors

As per Section 5.2 of [RFC5988], a link description MAY include an "anchor" parameter, in which case the context is the URI included in that attribute. This is used to describe a relationship between two resources. A consuming implementation can, however, choose to ignore such links. It is not expected that all implementations will be able to derive useful information from explicitly anchored links.

## 3. CoRE Link Attributes

The following CoRE-specific target attributes are defined in addition to those already defined in [RFC5988]. These attributes describe information useful in accessing the target link of the relation and, in some cases, can use the syntactical form of a URI. Such a URI MAY be dereferenced (for instance, to obtain a description of the link relation), but that is not part of the protocol and MUST NOT be done automatically on link evaluation. When the values of attributes are compared, they MUST be compared as strings.

### 3.1. Resource Type 'rt' Attribute

The Resource Type 'rt' attribute is an opaque string used to assign an application-specific semantic type to a resource. One can think of this as a noun describing the resource. In the case of a temperature resource, this could be, e.g., an application-specific semantic type like "outdoor-temperature" or a URI referencing a specific concept in an ontology like "http://sweet.jpl.nasa.gov/2.0/phys.owl#Temperature". Multiple Resource Types MAY be included in the value of this parameter, each separated by a space, similar to the relation attribute. The registry for Resource Type values is defined in Section 7.4.

The Resource Type attribute is not meant to be used to assign a human-readable name to a resource. The "title" attribute defined in [RFC5988] is meant for that purpose. The Resource Type attribute MUST NOT appear more than once in a link.

### 3.2. Interface Description 'if' Attribute

The Interface Description 'if' attribute is an opaque string used to provide a name or URI indicating a specific interface definition used to interact with the target resource. One can think of this as describing verbs usable on a resource. The Interface Description attribute is meant to describe the generic REST interface to interact with a resource or a set of resources. It is expected that an Interface Description will be reused by different Resource Types. For example, the Resource Types "outdoor-temperature", "dew-point", and "rel-humidity" could all be accessible using the Interface Description "http://www.example.org/myapp.wadl#sensor". Multiple Interface Descriptions MAY be included in the value of this parameter, each separated by a space, similar to the relation attribute. The registry for Interface Description values is defined in Section 7.4.

The Interface Description could be, for example, the URI of a Web Application Description Language (WADL) [WADL] definition of the target resource "http://www.example.org/myapp.wadl#sensor", a URN indicating the type of interface to the resource "urn:myapp:sensor", or an application-specific name "sensor". The Interface Description attribute MUST NOT appear more than once in a link.

### 3.3. Maximum Size Estimate 'sz' Attribute

The maximum size estimate attribute 'sz' gives an indication of the maximum size of the resource representation returned by performing a GET on the target URI. For links to CoAP resources, this attribute is not expected to be included for small resources that can comfortably be carried in a single Maximum Transmission Unit (MTU) but SHOULD be included for resources larger than that. The maximum size estimate attribute MUST NOT appear more than once in a link.

Note that there is no defined upper limit to the value of the 'sz' attributes. Implementations MUST be prepared to accept large values. One implementation strategy is to convert any value larger than a reasonable size limit for this implementation to a special value "Big", which in further processing would indicate that a size value was given that was so big that it cannot be processed by this implementation.

## 4. Well-Known Interface

Resource discovery in CoRE is accomplished through the use of a well-known resource URI that returns a list of links about resources hosted by that server and other link relations. Well-known resources

have a path component that begins with "/.well-known/" as specified in [RFC5785]. This specification defines a new well-known resource for CoRE Resource Discovery: "/.well-known/core".

A server implementing this specification MUST support this resource on the default port appropriate for the protocol for the purpose of resource discovery. It is, however, up to the application which links are included and how they are organized. The resource "/.well-known/core" is meant to be used to return links to the entry points of resource interfaces on a server. More sophisticated link organization can be achieved by including links to CoRE Link Format resources located elsewhere on the server, for example, to achieve an index. In the absence of any links, a zero-length payload is returned. The resource representation of this resource MUST be the CoRE Link Format described in Section 2.

The CoRE resource discovery interface supports the following interactions:

- Performing a GET on "/.well-known/core" to the default port returns a set of links available from the server (if any) in the CoRE Link Format. These links might describe resources hosted on that server or on other servers or express other kinds of link relations as described in Section 2.

- Filtering may be performed on any of the link format attributes using a query string as specified in Section 4.1. For example, [GET /.well-known/core?rt=temperature-c] would request resources with the Resource Type temperature-c. A server is not, however, required to support filtering.

- More capable servers such as proxies could support a resource directory by requesting the resource descriptions of other end-points or allowing servers to POST requests to "/.well-known/core". The details of such resource directory functionality is, however, out of the scope of this specification and is expected to be specified separately.

### 4.1. Query Filtering

A server implementing this specification MAY recognize the query part of a resource discovery URI as a filter on the resources to be returned. The path and query components together should conform to the following level-4 URI Template [RFC6570]:

```
    /.well-known/core{?search*}
```

where the variable "search" is a 1-element list that has a single name/value pair, where

- name is either "href", a link-param name defined in this specification, or any other link-extension name, and
- value is either a Complete Value String that does not end in an "*" (%2A), or a Prefix Value String followed by an "*" (%2A).

The search name "href" refers to the URI-reference between the "<" and ">" characters of a link. Both Value Strings match a target attribute only if it exists. Value Strings are percent-decoded ([RFC3986], Section 2.1) before matching; similarly, any target attributes notated as quoted-string are interpreted as defined in Section 2.2 of [RFC2616]. After these steps, a Complete Value String matches a target attribute if it is bitwise identical. A Prefix Value String matches a target attribute if it is a bitwise prefix of the target attribute (where any string is a prefix of itself). Empty Prefix Value Strings are allowed; by the definition above, they match any target attribute that does exist. Note that relation-type target attributes can contain multiple values, and each value MUST be treated as a separate target attribute when matching.

It is not expected that very constrained nodes support filtering. Implementations not supporting filtering MUST simply ignore the query string and return the whole resource for unicast requests.

When using a transfer protocol like the Constrained Application Protocol (CoAP) that supports multicast requests, special care needs to be taken. A multicast request with a query string SHOULD NOT be responded to if filtering is not supported or if the filter does not match (to avoid a needless response storm). The exception is in cases where the IP stack interface is not able to indicate that the destination address was multicast.

The following are examples of valid query URIs:

- ?href=/foo matches a link-value that is anchored at /foo
- ?href=/foo* matches a link-value that is anchored at a URI that starts with /foo
- ?foo=bar matches a link-value that has a target attribute named foo with the exact value bar
- ?foo=bar* matches a link-value that has a target attribute named foo, the value of which starts with bar, e.g., bar or barley
- ?foo=* matches a link-value that has a target attribute named foo

## 5. Examples

A few examples of typical link descriptions in this format follows. Multiple resource descriptions in a representation are separated by commas. Linefeeds are also included in these examples for readability. Although the following examples use CoAP response codes, the examples are applicable to HTTP as well (the corresponding response code would be 200 OK).

This example includes links to two different sensors sharing the same Interface Description. Note that the default relation type for this link format is "hosts" in links with no rel= target attribute. Thus, the links in this example tell that the Origin server from which /.well-known/core was requested (the context) hosts the resources /sensors/temp and /sensors/light (each a target).

```
REQ: GET /.well-known/core

RES: 2.05 Content
</sensors/temp>;if="sensor",
</sensors/light>;if="sensor"
```

Without the linefeeds inserted here for readability, the format actually looks as follows.

```
</sensors/temp>;if="sensor",</sensors/light>;if="sensor"
```

This example arranges link descriptions hierarchically, with the entry point including a link to a sub-resource containing links about the sensors.

```
REQ: GET /.well-known/core

RES: 2.05 Content
</sensors>;ct=40

REQ: GET /sensors

RES: 2.05 Content
</sensors/temp>;rt="temperature-c";if="sensor",
</sensors/light>;rt="light-lux";if="sensor"
```

An example query filter may look like:

```
REQ: GET /.well-known/core?rt=light-lux

RES: 2.05 Content
</sensors/light>;rt="light-lux";if="sensor"
```

Note that relation-type attributes like 'rt', 'if', and 'rel' can have multiple values separated by spaces. A query filter parameter can match any one of those values, as in this example:

```
REQ: GET /.well-known/core?rt=light-lux

RES: 2.05 Content
</sensors/light>;rt="light-lux core.sen-light";if="sensor"
```

This example shows the use of an "anchor" attribute to relate the temperature sensor resource to an external description and to an alternative URI.

```
REQ: GET /.well-known/core

RES: 2.05 Content
</sensors>;ct=40;title="Sensor Index",
</sensors/temp>;rt="temperature-c";if="sensor",
</sensors/light>;rt="light-lux";if="sensor",
<http://www.example.com/sensors/t123>;anchor="/sensors/temp"
;rel="describedby",
</t>;anchor="/sensors/temp";rel="alternate"
```

If a client is interested in finding relations about a particular resource, it can perform a query on the anchor parameter:

```
REQ: GET /.well-known/core?anchor=/sensors/temp

RES: 2.05 Content
<http://www.example.com/sensors/temp123>;anchor="/sensors/temp"
;rel="describedby",
</t>;anchor="/sensors/temp";rel="alternate"
```

The following example shows a large firmware resource with a size attribute. The consumer of this link would use the 'sz' attribute to determine if the resource representation is too large and if block transfer would be required to request it. In this case, a client with only a 64 KiB flash might only support a 16-bit integer for storing the 'sz' attribute. Thus, a special flag or value should be used to indicate "Big" (larger than 64 KiB).

```
REQ: GET /.well-known/core?rt=firmware

RES: 2.05 Content
</firmware/v2.1>;rt="firmware";sz=262144
```

## 6. Security Considerations

This specification has the same security considerations as described in Section 7 of [RFC5988]. The "/.well-known/core" resource MAY be protected, e.g., using Datagram Transport Layer Security (DTLS) when hosted on a CoAP server as per [COAP], Section 9.1.

Some servers might provide resource discovery services to a mix of clients that are trusted to different levels. For example, a lighting control system might allow any client to read state variables, but only certain clients to write state (turn lights on or off). Servers that have authentication and authorization features SHOULD support authentication features of the underlying transport protocols (HTTP or DTLS/TLS) and allow servers to return different lists of links based on a client's identity and authorization. While such servers might not return all links to all requesters, not providing the link does not, by itself, control access to the relevant resource -- a bad actor could know or guess the right URIs. Servers can also lie about the resources available. If it is important for a client to only get information from a known source, then that source needs to be authenticated.

Multicast requests using CoAP for the well-known link-format resources could be used to perform denial of service on a constrained network. A multicast request SHOULD only be accepted if the request is sufficiently authenticated and secured using, e.g., IPsec or an appropriate object security mechanism.

CoRE Link Format parsers should be aware that a link description may be cyclical, i.e., contain a link to itself. These cyclical links could be direct or indirect (i.e., through referenced link resources). Care should be taken when parsing link descriptions and accessing cyclical links.

## 7. IANA Considerations

### 7.1. Well-Known 'core' URI

This memo registers the 'core' well-known URI in the Well-Known URIs registry as defined by [RFC5785].

URI suffix: core

Change controller: IETF

Specification document(s): RFC 6690

Related information: None

### 7.2. New 'hosts' Relation Type

This memo registers the new "hosts" Web Linking relation type as per [RFC5988].

Relation Name: hosts

Description: Refers to a resource hosted by the server indicated by the link context.

Reference: RFC 6690

Notes: This relation is used in CoRE where links are retrieved as a "/.well-known/core" resource representation and is the default relation type in the CoRE Link Format.

Application Data: None

### 7.3. New 'link-format' Internet Media Type

This memo registers the a new Internet media type for the CoRE Link Format, 'application/link-format'.

Type name: application

Subtype name: link-format

Required parameters: None

Optional parameters: None

Encoding considerations: Binary data (UTF-8)

Security considerations:

Multicast requests using CoAP for the well-known link-format resources could be used to perform denial of service on a constrained network. A multicast request SHOULD only be accepted if the request is sufficiently authenticated and secured using, e.g., IPsec or an appropriate object security mechanism.

CoRE Link Format parsers should be aware that a link description may be cyclical, i.e., contain a link to itself. These cyclical links could be direct or indirect (i.e., through referenced link resources). Care should be taken when parsing link descriptions and accessing cyclical links.

Interoperability considerations: None

Published specification: RFC 6690

Applications that use this media type: CoAP server and client implementations for resource discovery and HTTP applications that use the link-format as a payload.

Additional information:

Magic number(s):

File extension(s): *.wlnk

Macintosh file type code(s):

Intended usage: COMMON

Restrictions on usage: None

Author: CoRE WG

Change controller: IETF

### 7.4. Constrained RESTful Environments (CoRE) Parameters Registry

This specification establishes a new Constrained RESTful Environments (CoRE) Parameters registry, which contains two new sub-registries of Link Target Attribute values (defined in [RFC5988]), one for Resource Type (rt=) Link Target Attribute values and the other for Interface Description (if=) Link Target Attribute values. No initial entries are defined by this specification for either sub-registry.

For both sub-registries, values starting with the characters "core" are registered using the IETF Review registration policy [RFC5226]. All other values are registered using the Specification Required policy, which requires review by a designated expert appointed by the IESG or their delegate.

The designated expert will enforce the following requirements:

- Registration values MUST be related to the intended purpose of these attributes as described in Section 3.

- Registered values MUST conform to the ABNF reg-rel-type definition of Section 2, meaning that the value starts with a lowercase alphabetic character, followed by a sequence of lowercase alphabetic, numeric, ".", or "-" characters, and contains no white space.

- It is recommended that the period "." character be used for dividing name segments and that the dash "-" character be used for making a segment more readable. Example Interface Description values might be "core.batch" and "core.link-batch".

- URIs are reserved for free use as extension values for these attributes and MUST NOT be registered.

Registration requests consist of the completed registration template below, with the reference pointing to the required specification. To allow for the allocation of values prior to publication, the designated expert may approve registration once they are satisfied that a specification will be published.

Note that Link Target Attribute Values can be registered by third parties if the Designated Expert determines that an unregistered Link Target Attribute Value is widely deployed and not likely to be registered in a timely manner.

The registration template for both sub-registries is:

- Attribute Value:
- Description:
- Reference:
- Notes: [optional]

Registration requests should be sent to the core-parameters@ietf.org mailing list, marked clearly in the subject line (e.g., "NEW RESOURCE TYPE - example" to register an "example" relation type or "NEW INTERFACE DESCRIPTION - example" to register an "example" Interface Description).

Within at most 14 days of the request, the Designated Expert(s) will either approve or deny the registration request, communicating this decision to the review list and IANA. Denials should include an explanation and, if applicable, suggestions as to how to make the request successful.

Decisions (or lack thereof) made by the Designated Expert can be first appealed to Application Area Directors (contactable using the app-ads@tools.ietf.org email address or directly by looking up their email addresses on http://www.iesg.org/ website) and, if the appellant is not satisfied with the response, to the full IESG (using the iesg@ietf.org mailing list).

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